MANAV GARG

AI / ML Engineer · MLOps · End-to-End ML Systems

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PROFESSIONAL SUMMARY

AI/ML Engineer with 2+ years of hands-on experience designing, training, and deploying machine learning systems end-to-end — from raw data ingestion to production inference. Deep expertise in MLOps, LLM/RAG architectures, multimodal models, and cloud-native ML infrastructure on AWS. Proven track record of cutting pipeline runtimes by 80%+, reducing hallucinations by 70%+, and slashing infrastructure costs by 20–25%. Comfortable owning the full stack: model research, scalable deployment, monitoring, and business-facing analytics.

EXPERIENCE

Independent Consultant — AI/ML · **Vault22** · *Remote* — Oct 2025 – Present

- Reduced overall infrastructure and ML pipeline operating costs by 20–25% through workload restructuring, resource-efficient automation, and compute-optimized AWS configurations.
- Architected and deployed scalable, self-hosted model serving infrastructure on AWS, improving performance reliability and reducing per-inference compute overhead.
- Rebuilt end-to-end ETL pipelines and data lake architecture, resolving systemic failures and improving data integrity, consistency, and cross-team accessibility.
- Designed and shipped custom admin portals with KPI dashboards in Amazon QuickSight, enabling real-time metric tracking for business leadership and investor reporting.
- Optimized MongoDB storage collections and indexes, and refactored backend pipelines to eliminate errors and improve query throughput.

Associate Data Scientist · **IHX Pvt. Ltd (A Perfios Company)** · *Bangalore, Onsite* — Jan 2025 – Mar 2026

- Led applied research in tariff digitization — built end-to-end ML pipelines automating extraction and standardization of large-scale financial documents, achieving measurable gains in accuracy and processing efficiency.
- Reduced data processing time from 24–36 hours to under 5 hours by pioneering optimized preprocessing and post-processing pipelines, while improving downstream dataset quality.
- Engineered prompt pipelines that cut LLM hallucination rates by over 70% and delivered consistent structured outputs from general-purpose models — without fine-tuning.
- Designed and deployed scalable asynchronous ML inference systems supporting continuous experimentation and real-time production serving under strict latency and throughput constraints.
- Built modular Python libraries for internal research and production ML pipelines, enabling reproducible experimentation and faster iteration across teams.
- Conducted deep research on multimodal architectures integrating RAG + LLM frameworks with YOLO, Donut, and custom encoder–decoder models for complex document understanding.

Industrial Trainee — AI/ML · **Intel Corporation** · *Chennai, Onsite* — May 2024 – Aug 2024

- Built an end-to-end NLP pipeline using Vision Language Models and event-driven architecture to classify contract clauses, detect deviations from templates, and flag fraud — with full pre/post-processing stages.
- Applied semantically rich embeddings to extract fine-print signals for contract validation and anomaly detection, delivering precise metrics for legal and compliance analysis.

Full Stack Developer & ML Engineer · **Tisac Pvt. Ltd.** · *Remote* — May 2024 – Jun 2024

- Deployed Jupyter Docker Lab environments for enterprise-grade data science workflows; built a Docker API service to automate container creation for model training at scale.
- Configured TLS/SSL HTTPS subdomains per container, reverse proxies, Firebase Auth, and MongoDB storage — delivering a secure, production-ready MLOps infrastructure.

Full Stack Developer Intern · **Hogarth India (WPP)** · *Gurugram, Onsite* — Aug 2023 – Oct 2023

- Contributed to the Coca-Cola / Thums Up ICC World Cup 2023 campaign — built full-stack features for live vote logging, code redemption, and real-time polling using C# / .NET backend and optimized React frontend.

Android Developer & ML Intern · **Wipro** · *Chennai, Onsite* Aug 2023 – Nov 2023

- Built an ML-powered Android application for Wipro Infrastructure Engineering that used machine vision to detect and classify pathway obstructions for autonomous rover navigation.

TECHNICAL SKILLS

ML / AI: PyTorch, TensorFlow, Scikit-Learn, OpenCV, HuggingFace Transformers, Vision Transformers (ViT), YOLO, Donut, RAG, LLM Prompt Engineering, Multimodal Architectures, Encoder–Decoder Models
MLOps & Infra: Docker, Kubernetes (basics), AWS (EC2, S3, SageMaker, QuickSight), Asynchronous Inference, CI/CD Pipelines, Model Deployment & Serving, ETL Pipeline Design, Data Lake Architecture
Data & Analytics: Pandas, NumPy, Matplotlib, SQL, MongoDB, NoSQL, Amazon QuickSight, KPI Dashboard Design
Languages: Python, C/C++, JavaScript / TypeScript, Java, SQL, Bash, Dart, HTML5/CSS3, LaTeX
Web & Backend: FastAPI, Django, Flask, Node.js / Express, React / Next.js, REST APIs, Firebase, Vercel
Tools & Platforms: Git / GitHub, Linux / Unix, VS Code, Vim, Figma, Adobe Photoshop / Illustrator

PROJECTS

LuminSkin: Skin Cancer Classification using Vision Transformers | *Python, PyTorch, ViT, DeepViT*

- Trained and benchmarked standard ViT vs. DeepViT for medical image classification, achieving state-of-the-art accuracy on skin cancer datasets and advancing early-detection tooling.

Satellite Image Analysis using Vision Transformers | *Python, PyTorch, ViT, CCT*

- Achieved SOTA in land-use classification by comparing ViT vs. Compact Convolutional Transformer (CCT), enabling satellite-based environmental monitoring and urban planning tools.

KeyKeg — Password Strength Analyzer & Manager | *Python, Scikit-Learn, Random Forest, Next.js*

- Built a full-stack password management app with ML-powered strength classification (Random Forest), secure storage, and strong password generation — complete end-to-end integration.

NSCHR — Web HMIS Portal | *Express, EJS, MongoDB, CSS3*

- Designed and sold a Health Management Information System (HMIS) portal to local clinics, enabling fast patient record access for medical practitioners at point of care.

EDUCATION

B.Tech, Computer Science Engineering Graduated Jun 2025

Vellore Institute of Technology, Chennai · *Specialization: Artificial Intelligence & Machine Learning* · *CGPA: 8.51 / 10*

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Artificial Intelligence, XAI, Linear Algebra, Probability & Statistics, OOP